European Commission

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Artificial Intelligence

Updated report 2025

Data Management Working Group

October 2025

Artificial Intelligence

Data Management Working Group

October 2025

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ACKNOWLEDGEMENTS

The editors would like to acknowledge the valuable inputs from BRIDGE projects and Data Management WG who participated in the meetings and surveys as well as members of the Bridge secretariat team, all contributing to this BRIDGE Artificial Intelligence Report.

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PDF Web ISBN: 978-92-9405-341-1 doi: 10.2926/7351215 HZ-01-25-155-EN-N

Luxembourg: Publications Office of the European Union, 2025

Manuscript completed in October 2025

First edition

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1 List of Acronyms and Abbreviations

AI	Artificial Intelligence
BPMN	Business Process Model and Notation
CEN	European Committee for Standardization
CENELEC	European Committee for Electrotechnical Standardization
DSO	Distribution System Operator
EMS	Energy Management System
EU	European Union
EC	European Commission
DMWG	Data Management Working Group
IEC	International Electrotechnical Commission
KPI	Key Performance Indicator
LLM	Large Language Model
ML	Machine Learning
SCADA	Supervisory Control and Data Acquisition
SGAM	Smart Grid Architecture Model
SO	System Operator
TSO	Transmission System Operator
TRL	Technology Readiness Level
UML	Unified Modeling Language
WG	Working Group

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Executive Summary

The *Action on Artificial Intelligence (AI)* was launched in April 2024 under the BRIDGE Data Management Working Group, in response to the increasing relevance of AI technologies in supporting the digitalisation and decarbonisation of Europe's energy systems. This Action aimed to map existing AI applications across BRIDGE projects, assess their maturity, and understand both the opportunities and challenges encountered in deploying AI-based solutions.

A structured survey involving over 30 BRIDGE projects gathered detailed information on AI use cases, the types of data used, maturity levels (based on technology readiness levels), benefits compared to conventional approaches, and barriers to development and deployment. Complementary group discussions enriched the analysis with project-specific insights and emerging themes.

The results showed that AI is already being applied in diverse areas such as forecasting, predictive maintenance, grid optimisation, and customer analytics. Most applications are in the design or pilot stage (TRL 4–7), indicating that BRIDGE projects are moving beyond experimentation into real-world validation. Reported benefits include improved decision-making, enhanced efficiency, and cost reductions. However, significant challenges remain, particularly in relation to data quality and availability, interoperability, system integration, and AI model explainability.

The Action also revealed growing awareness among projects of the need for trustworthy and responsible AI. Many respondents reported addressing ethical, environmental, and societal implications, and expressed a desire for shared guidance on governance and regulatory compliance — including the emerging EU AI Act.

This report consolidates the key insights from the survey, offers a set of practical recommendations for BRIDGE, project consortia, and the European Commission, and proposes a roadmap for future activities. These include establishing a dedicated AI subgroup, deepening the mapping of AI applications, aligning with European data spaces and standardisation efforts, and developing common methodologies for documenting and evaluating AI in energy contexts.

As a result, the Action on AI lays the groundwork for ongoing collaboration and positions BRIDGE to take a leading role in advancing the responsible, interoperable, and impactful integration of AI into Europe's clean energy transition.

Introduction

The “Action on AI” was proposed during the BRIDGE General Assembly held in April 2024 [1], with the aim of gathering insights on how AI is currently being used or developed within BRIDGE projects. The initial focus was to map existing and planned AI applications, in order to better understand what has been achieved so far, the maturity of available solutions, and the experience gained by projects in deploying AI-based approaches. This included identifying the main challenges encountered, such as data availability, interoperability, and trustworthiness of AI models, as well as highlighting the opportunities and added value that AI can bring to future developments in the energy domain.

To this end, a collaborative effort was launched among several Horizon 2020 and Horizon Europe projects participating in BRIDGE, with the objective of consolidating knowledge, use cases, and technical insights related to AI applications across different parts of the energy system.

In 2024–2025, the activities of the Action on AI have focused on three main topics:

- **Mapping AI use cases** within BRIDGE projects, with a focus on their application to data-driven tasks such as forecasting, optimisation, pattern recognition, and anomaly detection
- **Assessing the maturity and impact** of AI solutions, including the challenges faced during their development or deployment, and the conditions required to support their adoption
- **Providing recommendations** to improve data management frameworks and practices within BRIDGE, so they can better support the integration of AI in energy sector applications

This report presents the outcomes of these activities, offering a consolidated view of AI adoption across BRIDGE projects, and suggesting directions for future work in support of trustworthy and effective AI-enabled energy systems.

2 General approach

2.1 Context

The **Action on AI** was initiated within the BRIDGE Data Management Working Group to gain a clearer understanding of how **artificial intelligence (AI)** is currently being used or developed in EU-funded projects across the energy sector. The initiative reflects the growing importance of AI technologies — such as machine learning, deep learning, reinforcement learning, and advanced data analytics — as **critical enablers of the digitalisation and decarbonisation** of energy systems.

In recent years, AI has emerged as a transformative tool across a wide range of **energy applications**, including:

- **Forecasting** of load, generation, and renewable energy production
- **Grid operation and optimisation**, supporting real-time control, congestion management, and flexibility activation
- **Predictive maintenance** of physical assets and infrastructure
- **Customer profiling and engagement**, enabling tailored demand response and efficiency programs
- **Energy trading and market analytics**, including portfolio optimisation, price prediction, and real-time bidding
- **Cybersecurity and anomaly detection**, enhancing situational awareness and response capability

These developments are fuelled by advances in computing infrastructure, the availability of granular real-time data, and a growing ecosystem of open-source and commercial AI tools. However, the deployment of AI in critical infrastructure also raises **important challenges**: access to high-quality and interoperable data, integration with legacy systems, lack of transparency in model decision-making, and broader concerns around **governance, ethics, and trust**.

In response to this context, the BRIDGE General Assembly held in April 2024 [1] endorsed the launch of the Action on AI, aiming to:

- Identify which AI applications are currently being developed or used in BRIDGE projects
- Assess their **maturity**, **benefits**, and **limitations**
- Understand the **data and infrastructure requirements** for AI deployment
- Collect experience on **challenges and success factors** from project implementation
- Provide practical **recommendations** for future collaboration and framework development

The Action complements and builds upon strategic initiatives such as the **European Commission *Digitalisation of Energy Action Plan*** [2]. This plan places AI at the heart of the EU's digital energy transition, emphasising the need for standardized and trustworthy AI systems. In line with this, recent global studies and strategic initiatives show how AI can improve the reliability and efficiency of energy systems, also highlighting practical applications in areas like utility operations, asset management, predictive analytics, and customer service [3]. At the same time, the plan addresses key challenges—such as operational, regulatory, and ethical issues—that must be overcome to scale AI effectively.

In addition, several Horizon Europe and Horizon 2020 projects have been actively applying AI to energy forecasting, grid flexibility coordination, market design, and energy data sharing, among others.

The Action on AI contributes to this growing body of knowledge by consolidating practical insights from BRIDGE projects, identifying shared barriers and enablers.

3 Methodological Approach

3.1.1 Call for participation

The first step in the process was to launch a **call for action** within the BRIDGE Data Management Working Group. This was intended to engage interested project representatives already active in BRIDGE and to form a dedicated group of contributors to the Action. The call was open to all projects wishing to share their experience with AI in the context of data handling, forecasting, control, optimization, and other relevant applications.

Once interest had been confirmed, the group collectively reviewed and refined the goals of the Action and agreed on the first concrete steps to be taken.

3.1.2 Questionnaire design and dissemination

The core of the methodology was the development of a **questionnaire** aimed at gathering structured input from all interested BRIDGE projects. The questionnaire was designed collaboratively by the Action participants and focused on the following key areas:

- General description of the project and its objectives
- Description of relevant use cases involving AI
- Type of AI methods and tools being used (e.g. machine learning, deep learning, reinforcement learning)
- Maturity level of the AI applications (e.g. experimental, piloted, deployed)
- Purpose of the AI solution and its expected or demonstrated advantages over traditional approaches
- Data requirements (e.g. type, volume, granularity, quality)
- Main challenges in the **development** (e.g. algorithm design, training data availability)
- Main challenges in the **deployment** (e.g. integration with existing systems, explainability, trust, scalability)

The questionnaire served as a common format to collect insights across projects and enabled a comparative analysis of AI applications across a broad spectrum of energy-related contexts.

3.1.3 Group discussions and alignment

In parallel with the survey, group discussions were held among Action participants to align on methodology, validate the questionnaire content, and interpret initial findings. This helped clarify the nuances behind the collected inputs and identify shared concerns and recurring patterns.

The collaborative nature of the process ensured that the results were rooted in real project experience and reflected the diversity of AI implementations in the energy domain.

3.1.4 Analysis and synthesis

The responses from the participating projects were analysed to identify commonalities and differences across use cases, technologies, and maturity levels. Applications were categorised by function (e.g. forecasting, asset optimisation, user engagement) and by the role AI played in the data value chain.

The results of the analysis form the basis of this report. They are intended to support a better understanding of the practical adoption of AI in BRIDGE projects and to guide the evolution of data management practices and frameworks — including recommendations on how to better integrate AI considerations into tools such as the BRIDGE Reference Framework and the Use Case Repository.

4 Activities Carried Out

Following the launch of the Action on AI during the BRIDGE General Assembly in April 2024, a structured work plan was implemented to guide the activities of the group and ensure broad engagement from participating BRIDGE projects. The main vehicle for gathering information and insights was a detailed questionnaire, designed collaboratively by the action participants to address both technical and strategic aspects of AI deployment in energy use cases.

The activities of the Action unfolded in three main phases:

1. Engagement of participants
2. Survey design and dissemination
3. Collection and initial analysis

4.1 Engagement of Participants

A call for participation was issued to the BRIDGE Data Management Working Group in mid-2024, inviting interested representatives from Horizon 2020 and Horizon Europe projects to join the Action on AI. The response confirmed strong interest across the BRIDGE community, with participants representing a mix research and innovation projects. This diverse representation ensured a broad and realistic view of AI use across the sector.

Once the group was established, a series of coordination meetings were held to set out the goals of the Action, prioritise topics of investigation, and co-design the data collection methodology. It was agreed that a structured survey would provide a practical and scalable way to gather input across multiple projects.

4.2 Survey Design and Dissemination

The questionnaire was developed through a collaborative and iterative process and structured into four sections:

- **Section 1: Personal and Organisational Information**: This part captured basic information about the respondents, such as their role in the project, type of organisation, experience in the energy sector, and familiarity with AI. It helped contextualise the responses and assess the diversity of perspectives.

- **Section 2: AI Use Cases**: The core of the survey focused on gathering information about specific AI use cases within each project. Respondents were asked to:

    - Identify the domain of application (e.g., forecasting, optimisation, asset management)

    - Describe the most relevant use case

    - Indicate the maturity of the AI solution, mapped to technology readiness levels (TRLs)

    - Specify data sources used (e.g., smart meters, weather, SCADA, market data)

    - Identify the AI techniques applied and the functional purpose (e.g., automation, decision support)

    - Highlight main development and deployment challenges (e.g., data quality, interoperability, system integration)

- **Section 3: Benefits and Opportunities:** This section explored the practical value of AI in improving operations, compared to traditional methods. Respondents identified perceived benefits such as increased efficiency, accuracy, and cost savings. They were also invited to share concrete examples of how AI added value and to reflect on potential future applications.

- **Section 4: Implementation Challenges and Trustworthiness:** The final part addressed barriers to implementation, including technical (e.g., model training, scalability), organisational (e.g., lack of expertise), and governance-related (e.g., transparency, ethics, regulatory constraints). The survey also inquired whether projects are actively considering ethical, societal, environmental, and trustworthiness aspects of their AI systems.

The survey was distributed among BRIDGE projects through WG channels, with a response period that allowed time for internal coordination within each consortium.

4.3 Collection and Initial Analysis

The Action on AI gathered responses through a structured survey distributed among BRIDGE projects. A total of **34 individual responses** were collected, representing a diverse set of organisations.

The responses covered a wide range of AI-related topics and were processed into structured categories to support both statistical analysis and thematic interpretation.

A first analysis of the maturity of AI solutions indicated the following distribution:

- **44%** of AI applications were at the **Design/Prototyping Stage (TRL 4–5)**
- **41%** at the **Deployment/Pilot Stage (TRL 6–7)**
- Only **15%** were still at the **Conceptual Stage (TRL 1–3)**

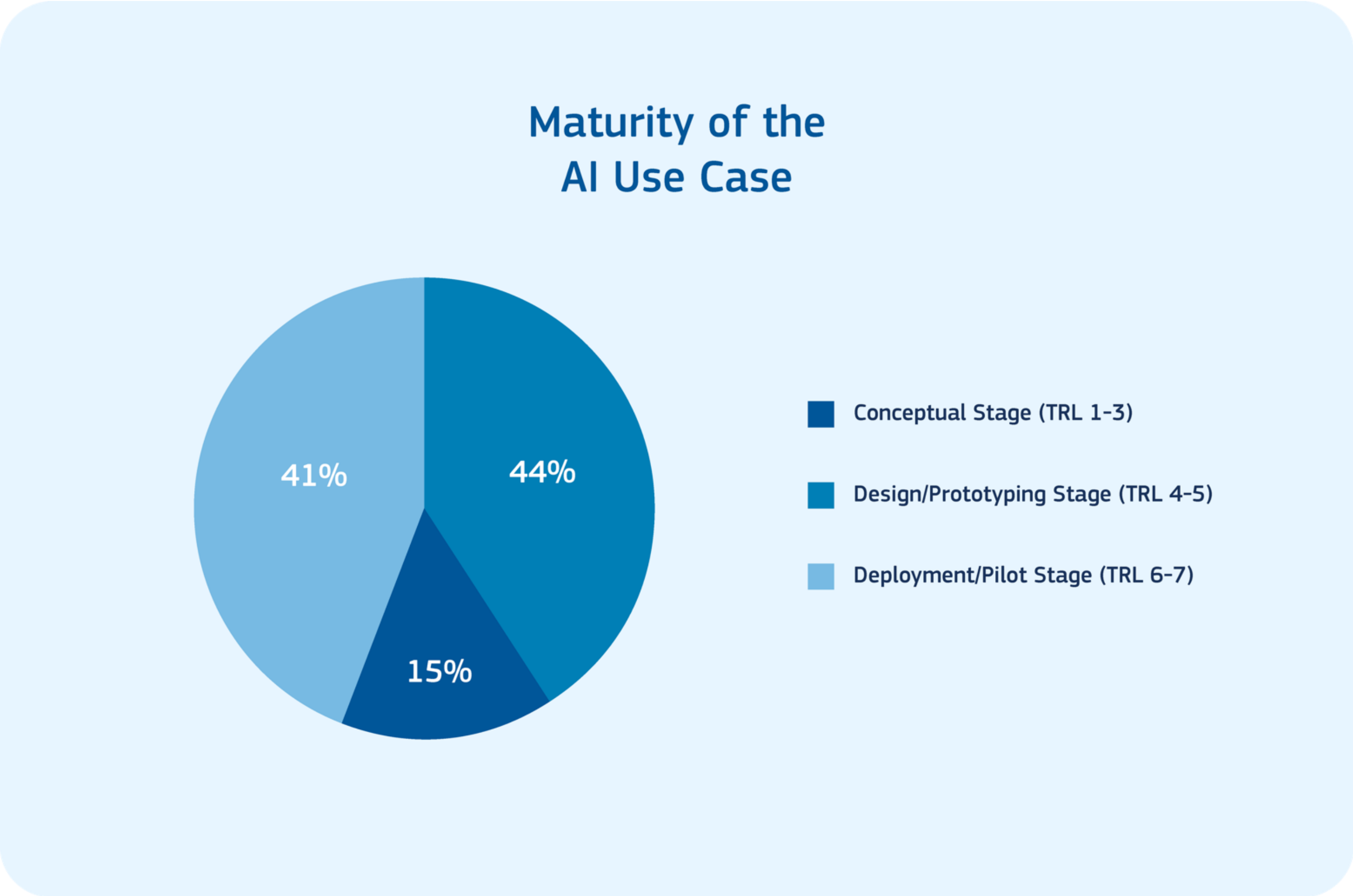

Figure 1 - Maturity of AI Use Cases participating in the questionnaire

This indicates that most projects are actively working on tangible AI solutions, many of which are being tested in real environments, although relatively few have reached full-scale operational deployment.

AI use cases were primarily focused on:

- **Decision support** (32%)
- **Data analysis** (28%)
- **Automation** (24%)
- **(Synthetic) data generation** (16%)

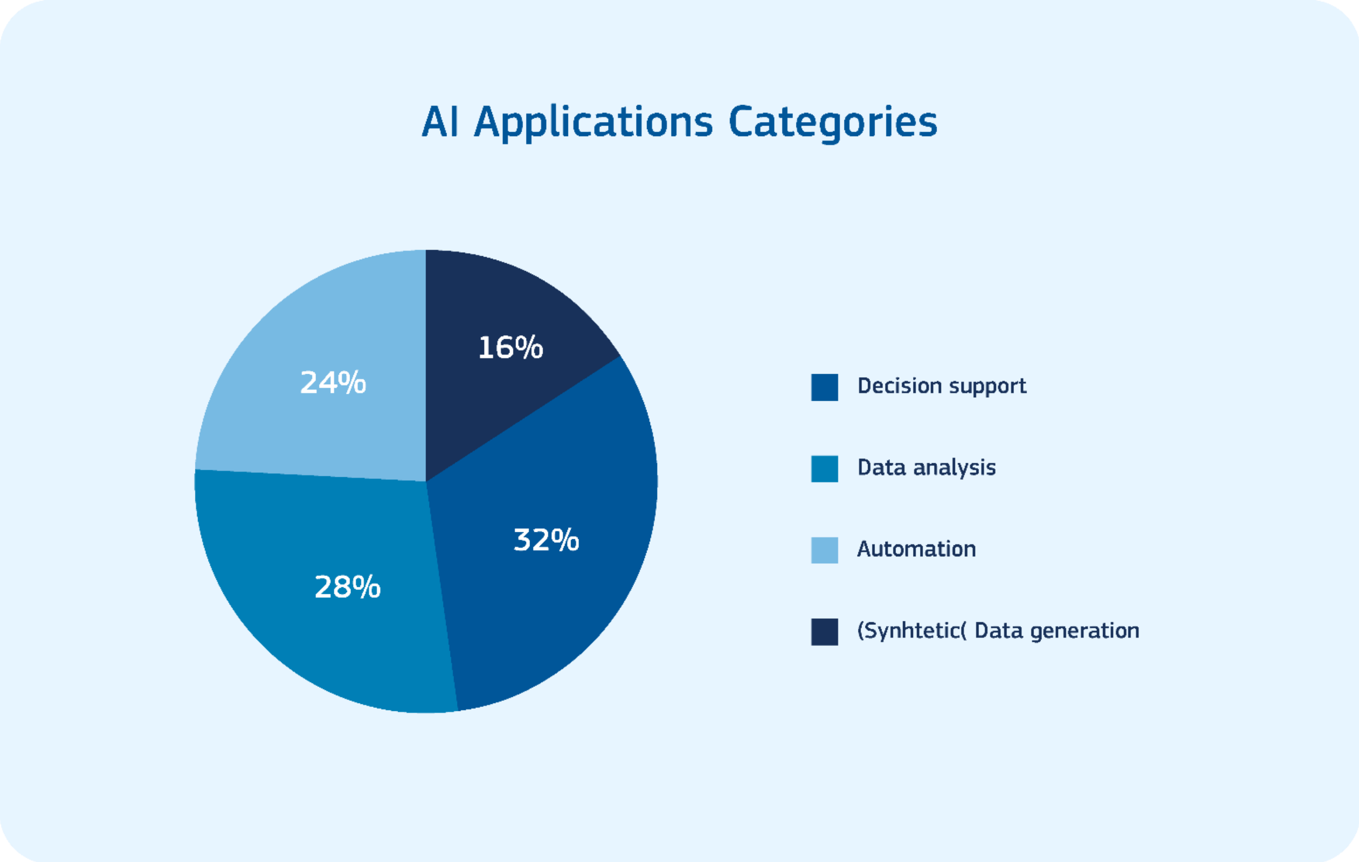

Figure 2 – Macro categories of AI applications

Regarding data requirements, the most commonly cited sources were:

- **Smart meter data**
- **Weather data**
- **Grid measurements** (voltage, current, phasors)
- **Market data**
- **Geospatial information**

The complete list of data requirements and related percentages is reported below.

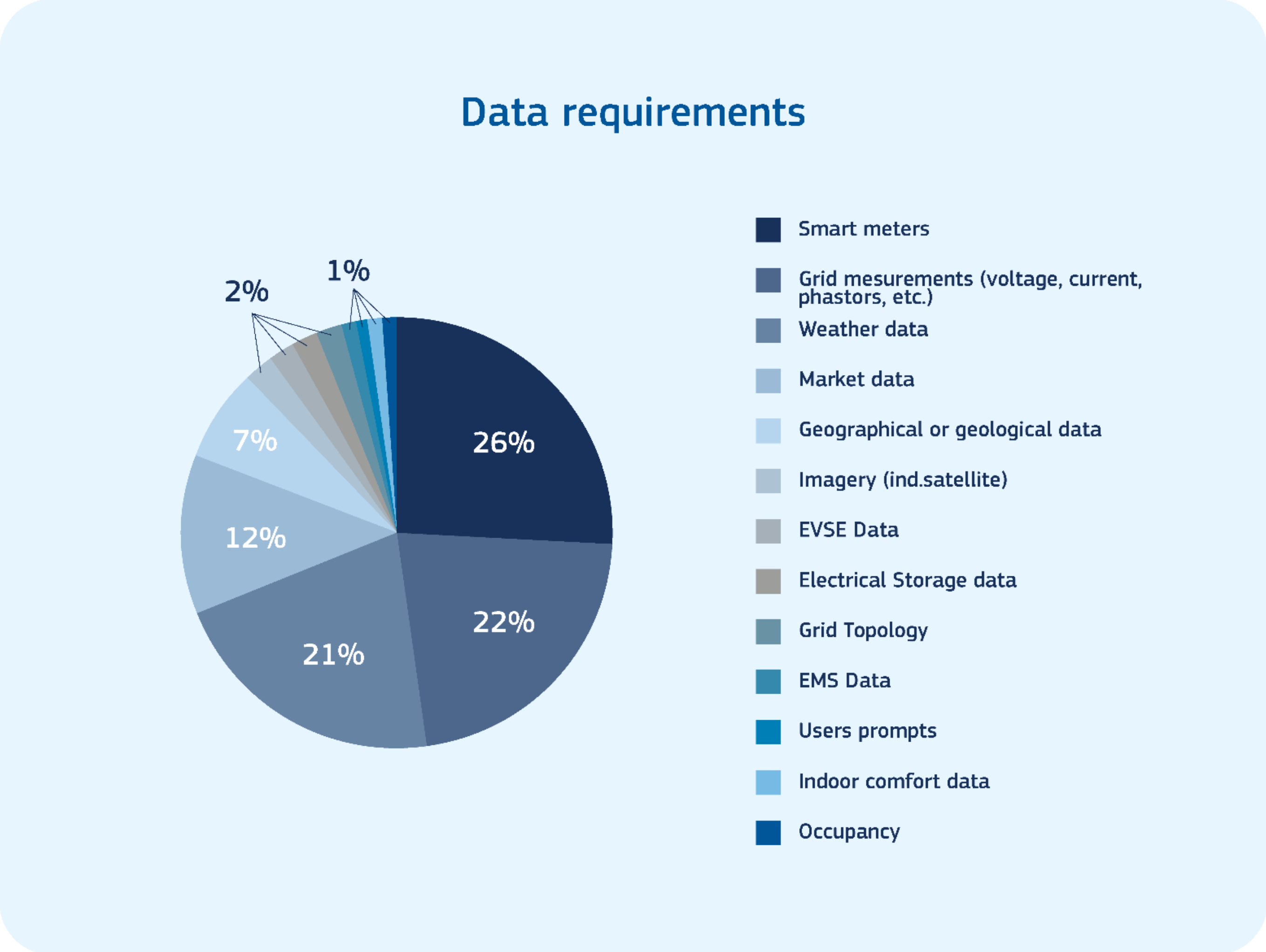

Figure 3 – Data input types

With respect to the challenges and benefits of AI applications, the participants' responses identified the following options as the most relevant.

Challenges were a recurring theme. The top implementation barriers were:

- **Data availability** (18.6%)
- **Data quality** (15.7%)
- **Model training** (10.4%)
- **AI transparency and explainability** (9%)
- **Data accuracy** (7.5%)

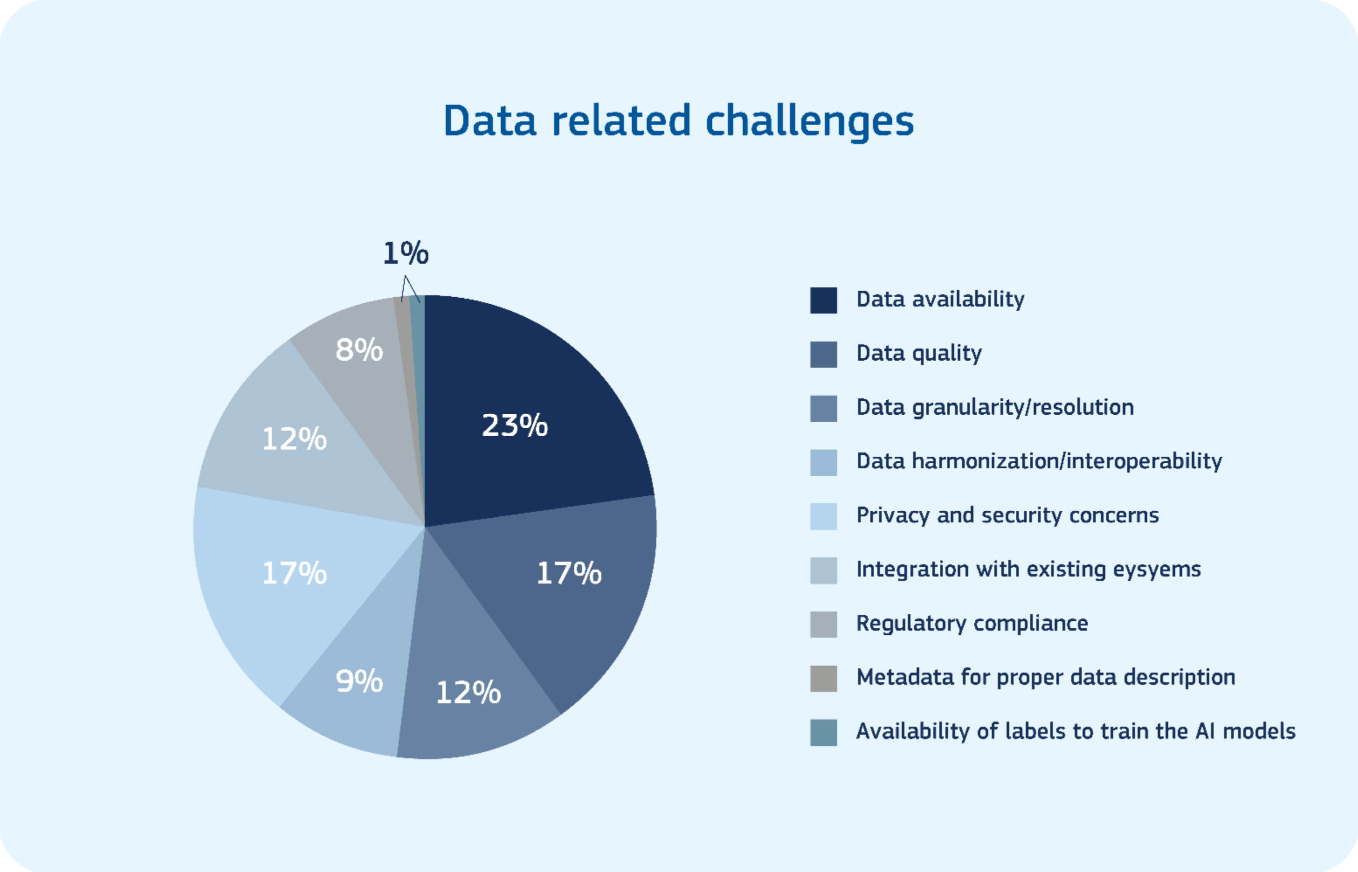

Figure 4 – Implementation challenges for AI applications

Nonetheless, projects also reported concrete benefits. The main improvements enabled by AI were:

- **Enhanced decision-making** (32.4%)
- **Increased efficiency** (29.4%)
- **Improved accuracy** (25%)
- **Cost reduction** (13.2%)

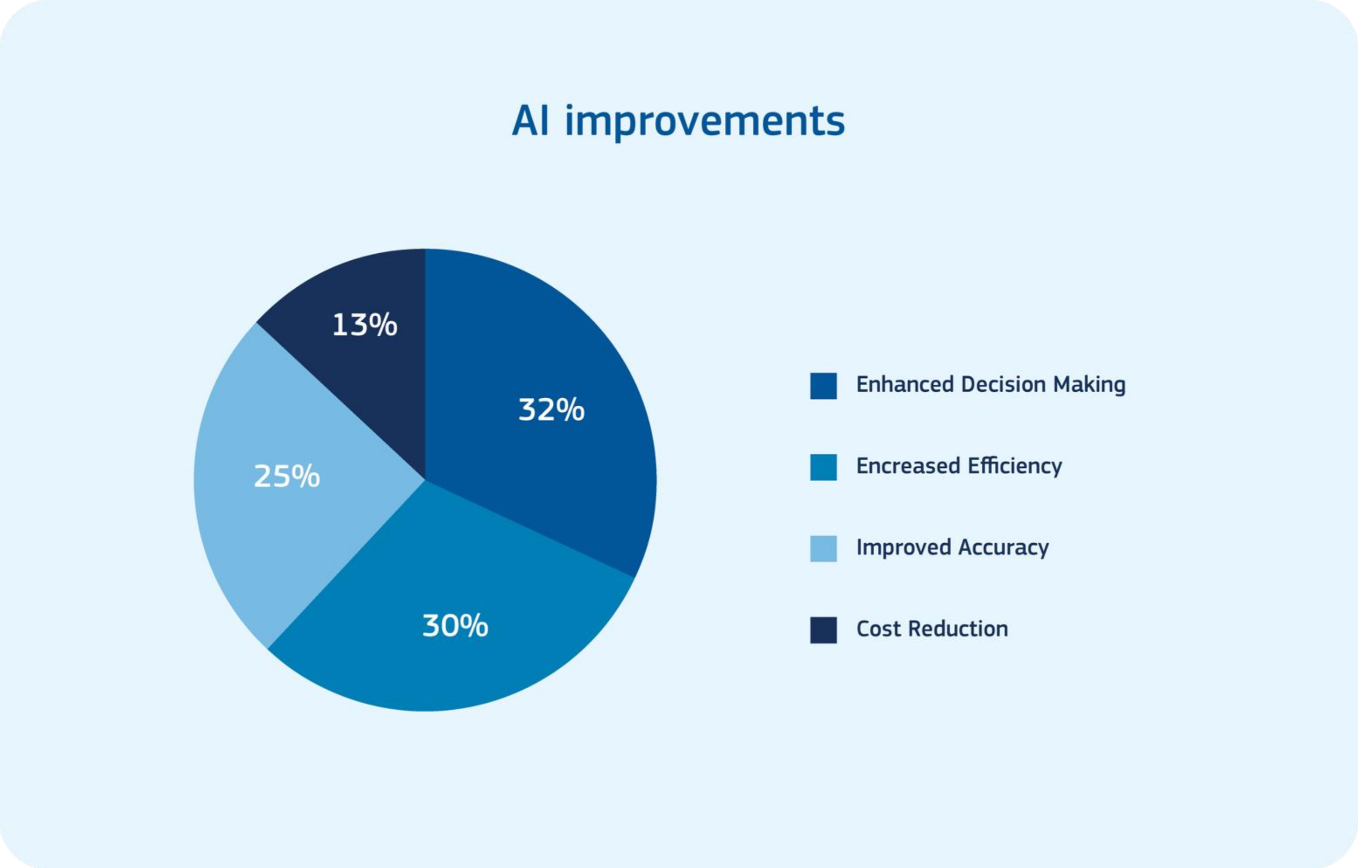

Figure 5 – Advantages offered by AI tools

These findings provided the foundation for further discussions and the development of consolidated insights, as presented in the next chapter.

In addition to technical and operational insights, the survey also explored whether projects were proactively addressing the **broader implications** of AI deployment. Specifically, respondents were asked whether their organisation or project is considering:

- **Ethical and societal impacts**
- **Environmental impacts**
- **Trustworthiness** of AI systems

The responses showed a **notable level of awareness** and engagement:

- **63.6%** of projects reported actively considering **ethical and societal impacts**
- **69.7%** indicated they are addressing **environmental impacts**
- **75.8%** confirmed that **trustworthiness** — including aspects such as transparency, reliability, and human oversight — is part of their AI development process

These figures demonstrate that while implementation challenges remain, there is a growing culture of **responsible AI development** within BRIDGE projects. This aligns with broader EU policy objectives, such as the **AI Act** and the emphasis on **trustworthy AI** in critical sectors like energy.

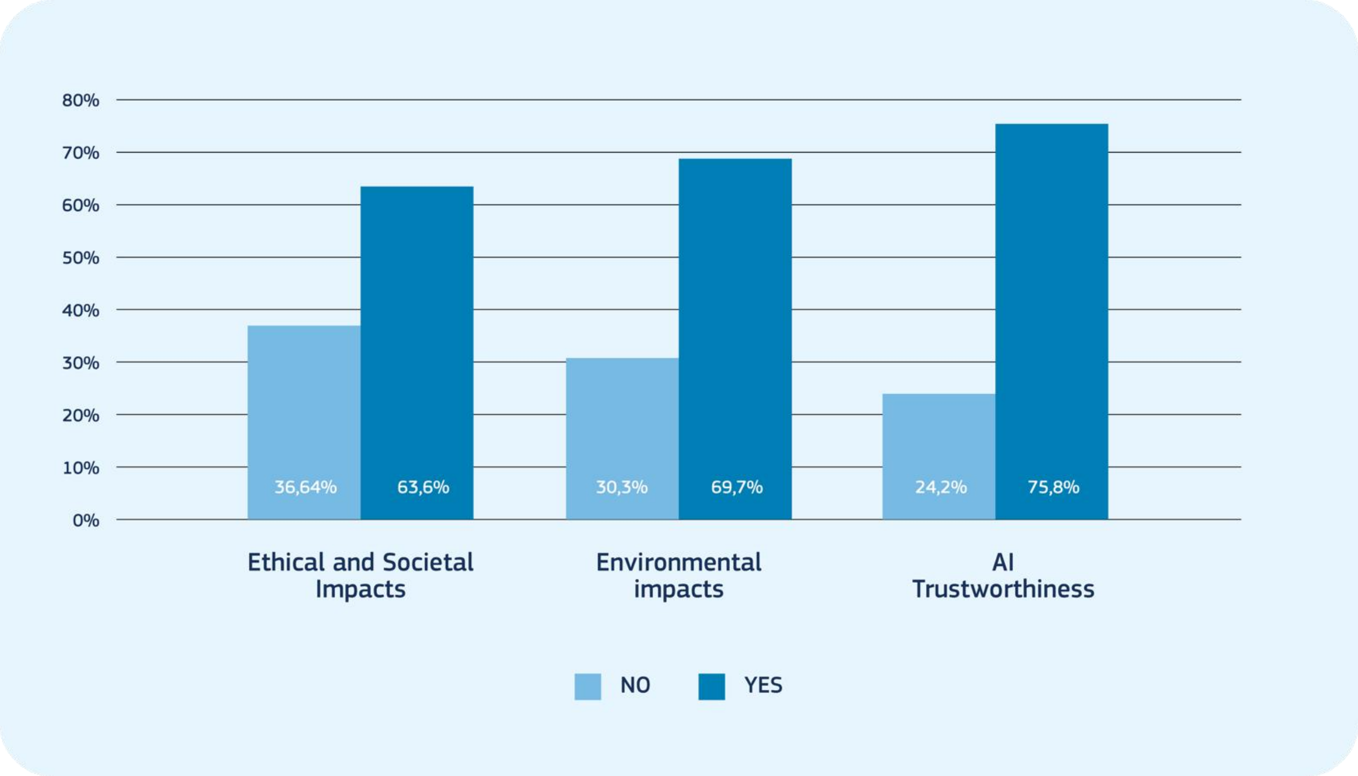

Figure 6 – Ethical, Environmental and Trustworthiness challenges

5 Key Insights

The analysis of the survey responses and complementary discussions with participating BRIDGE projects revealed several recurring themes, insights, and open challenges related to the use of AI in the energy sector. These findings provide a valuable snapshot of the current state of AI development within Horizon 2020 and Horizon Europe energy projects and serve to guide future collaboration and framework improvements within BRIDGE.

5.1 Diverse and Emerging AI Applications

The projects surveyed reported a wide range of AI applications across various domains of the energy system. The most frequently cited use cases included:

- **Forecasting**: of energy demand, renewable generation (especially solar and wind), and market prices.
- **Optimisation**: of grid operations, energy consumption, or flexibility resources.
- **Predictive maintenance**: for grid and asset management.
- **Customer analytics and engagement**: including segmentation, profiling, and behaviour prediction.
- **Anomaly detection and fault diagnostics**: in operational monitoring, cybersecurity, and automation systems.

This diversity reflects the increasing penetration of AI into multiple layers of the energy value chain, from operational control to customer interaction.

5.2 AI Maturity Still in Progress

Most AI use cases reported were in the **design/prototyping (TRL 4–5)** or **deployment/pilot (TRL 6–7)** stages. Only a few projects indicated that their AI solutions had reached full integration (TRL 8–9). This suggests that while interest and experimentation are widespread, **AI maturity is still developing**, and many solutions are not yet ready for large-scale deployment.

The lower levels of maturity were often associated with challenges in data acquisition, model robustness, regulatory uncertainty, and integration into existing operational frameworks.

5.3 Data Availability and Quality as Persistent Barriers

Across all surveyed projects, **data-related issues** were the most frequently reported challenges to AI development and deployment. These included:

- **Limited availability** of high-resolution or real-time data
- **Heterogeneity and lack of harmonisation** across data sources and formats
- **Insufficient data quality**, especially for training AI models
- Concerns around **privacy, security**, and **regulatory compliance**

These challenges often limit the ability to build, validate, and scale AI models effectively. Projects also highlighted the need for better tools and standards to facilitate data sharing across stakeholders.

5.4 Integration and Interoperability Challenges

A common obstacle in moving from prototyping to deployment was the **integration of AI tools with existing systems**, such as SCADA, EMS, or customer platforms. Several projects noted the lack of interfaces and standardisation for embedding AI models into operational workflows, as well as the high effort required to adapt legacy systems to accommodate new digital solutions.

This finding stresses the importance of aligning AI use cases with ongoing work in the BRIDGE **Reference Framework** and **Use Case Repository**, in order to support interoperability and replicability.

5.5 Explainability and Trustworthiness Emerging as Key Concerns

As AI becomes more embedded in decision-making processes, concerns around **transparency**, **explainability**, and **trust** are becoming increasingly relevant. Several projects expressed caution about deploying AI models whose internal logic is difficult to interpret — particularly in critical applications like grid management or market participation.

Moreover, while most projects had not yet formalised ethical or societal impact assessments, a growing number reported awareness of the need to consider:

- **Ethical AI principles**
- **Environmental implications** of AI training and deployment
- **Human oversight** mechanisms
- **AI governance and responsibility**

These concerns point to the growing importance of **"responsible AI"** practices in the energy sector.

5.6 Clear Value Propositions Identified

Despite the challenges, many projects reported **concrete advantages** of AI over traditional approaches, including:

- Improved forecasting accuracy
- Faster and more granular decision-making
- Better fault detection and predictive capabilities
- Increased operational efficiency and cost savings

These reported benefits strengthen the case for continued investment in AI-based innovation, provided that enabling conditions — particularly around data and integration — are adequately addressed.

6 Next Steps and Future Activities

The **Action on AI** has served as a first step in coordinating knowledge and practices around AI deployment within BRIDGE. Its results provide a valuable baseline from which future efforts can build. Several concrete directions have emerged from the analysis and the internal discussions among action participants point toward both short-term actions and longer-term opportunities for strategic alignment. By continuing and expanding this work, BRIDGE can position itself at the forefront of **responsible AI deployment** in energy, ensuring that emerging solutions are **technically robust, operationally integrated, and socially aligned** with Europe's clean energy and digital ambitions.

The outcomes of the **Action on AI** confirm both the growing **momentum** and the complexity of integrating **artificial intelligence** into energy systems. While the results of the survey and group discussions have shed light on current practices, challenges, and benefits, they have also made clear that this is only the beginning of a longer journey.

A clear first priority is to **consolidate and disseminate** the knowledge already gathered. The results of the survey, including aggregated data and key insights, will be published in a format accessible to all **BRIDGE participants**. In parallel, an **executive summary** or **policy brief** will be prepared to engage external stakeholders, such as **regulators**, **standardisation bodies**, and **policymakers**. Presenting these outcomes at upcoming **BRIDGE meetings** and **sector events** will help raise visibility and foster continued dialogue.

Building on this foundation, there is strong interest in continuing AI-related work within all working groups in BRIDGE, potentially through the creation of a **dedicated subgroup** or **task force**. Such a structure could provide a space for **ongoing exchange**, regular updates to the **AI use case mapping**, and deeper thematic exploration of issues such as **trust**, **regulation**, and **explainability**. It was also agreed that the participants of the original survey should be re-engaged in **follow-up discussions**. These would allow for a better understanding of the **full extent and impact of the challenges** identified, and help flesh out the concrete **best practices** adopted by projects to overcome them.

Another important line of work will focus on improving **methodologies**. A proposal emerged to establish a **common template** for documenting AI use cases across BRIDGE. This would make it easier to compare solutions, map them to the **Reference Framework**, and understand their **maturity**, **data needs**, and **trust features**. Enhancing the visibility of AI components within the **BRIDGE Use Case Repository** and aligning terminology across working groups will also be a priority.

In terms of **regulation** and **ecosystem alignment**, the group recognised the need for a deeper collective understanding of the **EU AI Act**. As this legislation will have implications for how AI is developed, documented, and governed in energy applications, it was recommended that this be addressed jointly with other **BRIDGE working groups**. **Standardisation activities** were also identified as highly relevant, particularly the work of the **CEN/CENELEC group** on **AI Trustworthiness Frameworks** [4], which will shape future practices around **explainability**, **accountability**, and **robustness**.

Another key insight relates to **data availability** — repeatedly flagged as one of the biggest barriers to AI adoption. Participants highlighted the importance of exploring how the emerging concept of **Common European Data Spaces**, and in particular **Energy Data Spaces**, can be leveraged to facilitate access to **high-quality**, **interoperable data**. This would help bridge the gap between **research-driven AI development** and **scalable, operational use** in energy systems.

Finally, the results of the Action can contribute to shaping the broader **strategic research and innovation agenda** in Europe. The findings will be relevant for ongoing **Horizon Europe** programming, particularly in **Cluster 5 (Climate, Energy and Mobility)**, and should also inform digital policy directions under the **Digital Europe Programme**. More broadly, BRIDGE can play a unique role in shaping how **AI governance**, **trustworthiness**, and

digital infrastructure are addressed across sectors — helping to set out a **responsible** and **interoperable path** for AI in energy.

In conclusion, the work started through the Action on AI has sparked both **momentum** and **commitment**. There is clear value in continuing this effort — not only to deepen understanding but also to guide real progress in the deployment of **trustworthy**, **efficient**, and **impactful AI** across Europe's energy system.

7 Conclusion

The **Action on AI** has provided the first coordinated effort within BRIDGE to explore the current landscape of Artificial Intelligence applications across EU-funded energy projects. Initiated in response to the growing relevance of AI in energy system planning, operation, and customer interaction, the Action aimed to identify the state of practice, assess maturity levels, understand challenges, and derive recommendations to support more effective and trustworthy AI integration.

Through a structured survey and active collaboration among BRIDGE project representatives, the Action collected and analysed insights from over 30 participating projects. These responses revealed a diverse and rapidly evolving set of AI use cases — from forecasting and optimization to predictive maintenance and customer engagement. While most solutions remain in the design or pilot stage (TRL 4–7), the survey showed clear momentum toward deployment and tangible benefits in terms of efficiency, accuracy, and decision support.

At the same time, persistent challenges were highlighted, particularly related to **data access and quality**, **system integration**, and **AI transparency and explainability**. These findings underscore the need for continued collaboration, shared methodologies, and support for responsible AI practices within the energy sector.

The Action on AI has produced several key outputs:

- A **comprehensive survey and analysis** of AI use cases in BRIDGE projects
- A summary of **common challenges and enablers**
- A set of **practical recommendations** for BRIDGE, project consortia, and policymakers
- A proposal for **future activities** to consolidate, expand, and align AI-related work within BRIDGE and with European digital and energy strategies

Looking ahead, the Action has laid the foundation for continued engagement on AI topics in BRIDGE. As AI continues to evolve and mature, its impact on the digital energy transition will grow — and so too will the need for **shared knowledge, harmonised frameworks**, and **collective governance**.

By building on the outcomes of this Action, the BRIDGE community is well-positioned to lead the development and deployment of **trustworthy, interoperable, and high-impact AI solutions** in support of Europe's energy and climate goals.

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The Service Contract (n. CINEA/2023/OP/0001/SI2.901723) supports ETIP SNET activities, funded by the EU.